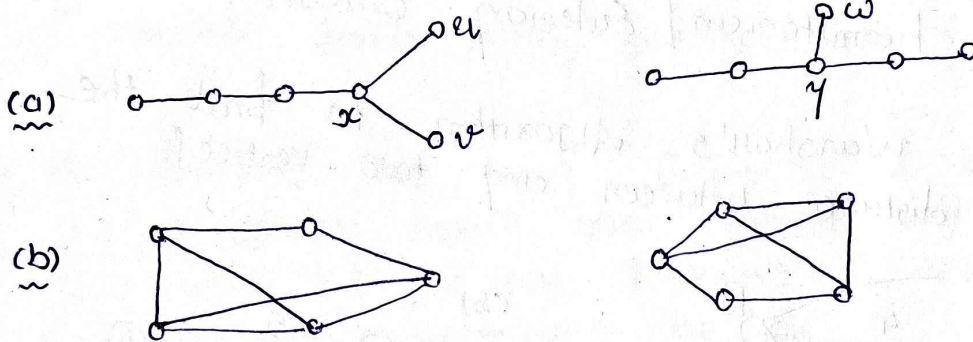


Unit-5 GraphsSEM-4

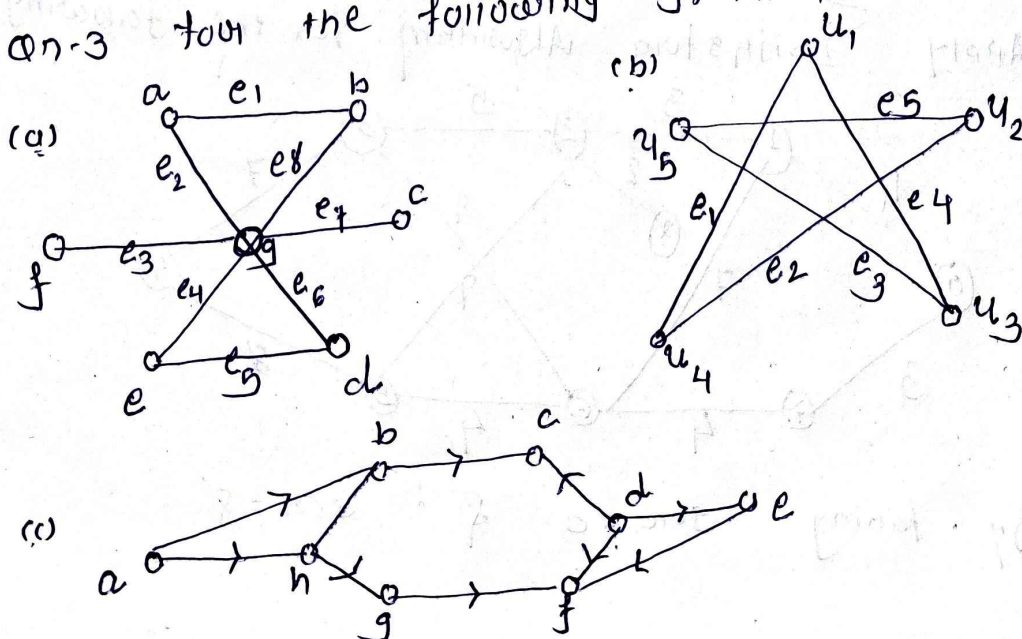
Qn-1 Define the following term with an example
Path, Circuit, Isolated vertex, Null graph,
Isomorphic graphs, Cut vertex, Simple graph,
Multi graph, Weighted graph, Degree of a vertex,
Connected graph, Multiple edges, Pseudo graph,
Subgraph of a graph.

Qn-2 check whether the following graphs are
isomorphic or not. Justify your conclusion.

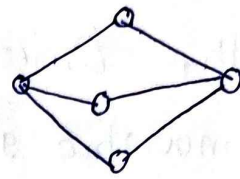
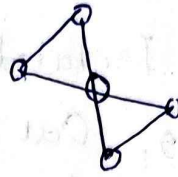
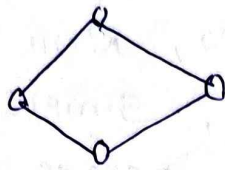
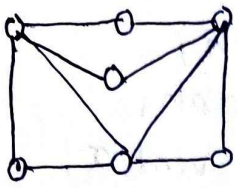


Qn-3 Define (i) Adjacency matrix (ii) Incidence
matrix and (iii) path matrix of a graph.

Qn-4 Find the associated matrices mentioned in
Qn-3 for the following graphs.



Qn-5: Define Euler graph & Hamiltonian graph with one example. Also identify which of the below are Eulerian or Hamiltonian.

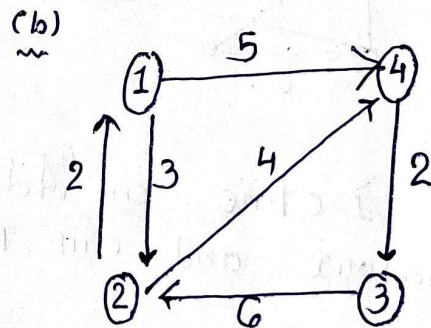
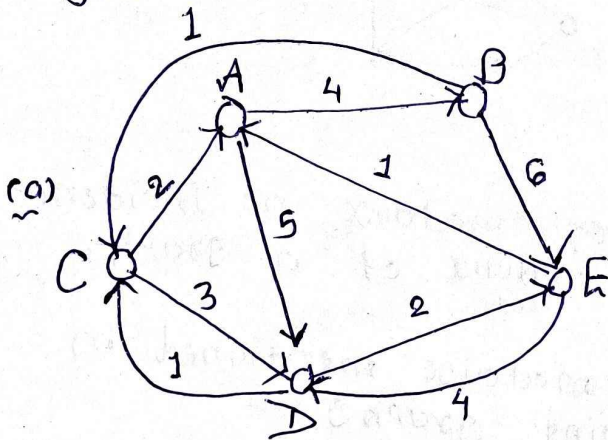


Q-6 Questions based on Euler / Hamiltonian graphs.

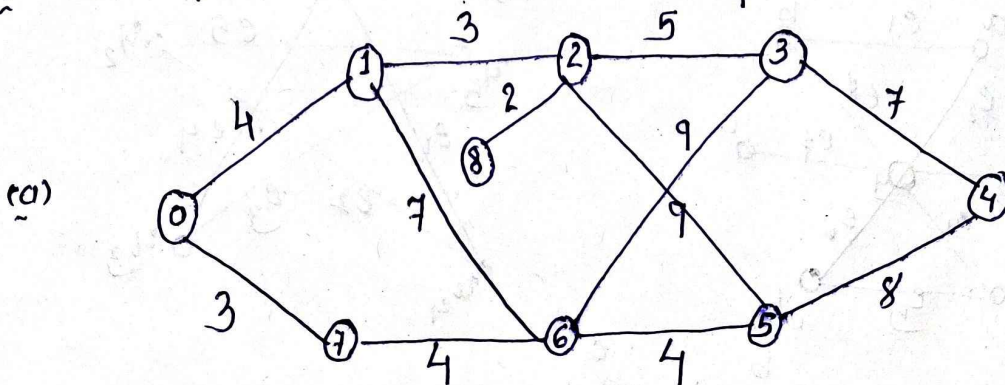
(1) A graph with vertices $\{A, B, C, D, E\}$ and edges $\{AB, BC, CD, DE, AE, AC\}$ is Eulerian or Hamiltonian. Justify.

(2) Does the Complete Bipartite graph $K_{3,3}$ is having Hamiltonian / Eulerian Circuit?

Q-7: Apply Warshall's Algorithm to find the shortest distance between any two vertices.

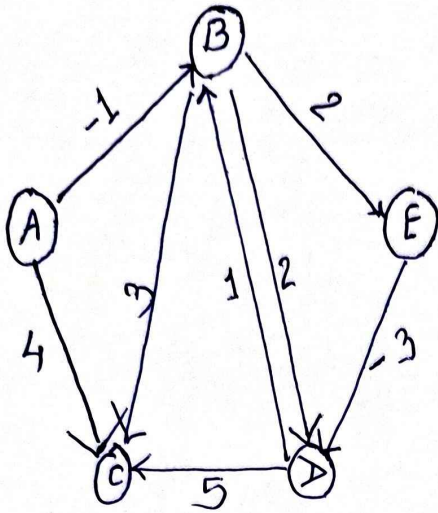


Q-8 Apply Dijkstra Algorithm for the following.



By taking $src = 0$ & $sct = 8$

Qn-9: Apply Bellman-Ford Algorithm



By taking $gnc = A$
& then $gnc = E$.

Qn-10 Define Regular graph, Complete graph & Connected graph with one example for each.

Qn-11: Explain the term Geodesic & distance in a graph with one example with at least 4 vertices & at least 7 edges by assigning random weights on edges with nos. 1 to 7 (can be different).

Qn-12: Give an example (Draw a graph) which is (1) Eulerian but not Hamiltonian & (2) Hamiltonian but not Eulerian.

~~~~~ The End ~~~~~